

A literature-implied answer to Tokarev's singleton-spectrum problem

Automated mathematical discovery run `fa_banach_001`, lane 7

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Source question. Tokarev [1] defines the ℓ_p -spectrum of an infinite-dimensional Banach space X by

$$S(X) = \{p : \ell_p \text{ is finitely representable in } X\}.$$

His Problem 2 asks whether there is a non-superreflexive Banach space X with $S(X) = \{2\}$. This is stated as a weaker companion to Problem 1, which asks for a non-superreflexive space with the hereditary approximation property (HAP).

Later input. Johnson and Szankowski [2, p. 1999, immediately after Question 3] recall that there exist nonreflexive Banach spaces of type $2 - \varepsilon$ and cotype $2 + \varepsilon$ for every $\varepsilon > 0$, citing the construction of Pisier and Xu [3]. The following standard argument identifies this as a positive answer to Tokarev's Problem 2.

Theorem 1. *Let X be an infinite-dimensional Banach space such that*

1. *X has type r for every $1 \leq r < 2$, and*
2. *X has cotype s for every $2 < s < \infty$.*

Then $S(X) = \{2\}$. Consequently, any nonreflexive Pisier–Xu space recalled in [2] is a non-superreflexive positive example for Tokarev's Problem 2.

Proof. Dvoretzky's theorem says that ℓ_2 is finitely representable in every infinite-dimensional Banach space, so $2 \in S(X)$.

Suppose first that $1 \leq p < 2$ and ℓ_p were finitely representable in X . Choose r with $p < r < 2$. A type- r inequality is a uniform finite-dimensional inequality, so it passes to every space finitely representable in X . Hence ℓ_p would have type r . This is impossible: the supremal type exponent of ℓ_p is p .

Now suppose that $2 < p < \infty$ and ℓ_p were finitely representable in X . Choose s with $2 < s < p$. Cotype- s inequalities likewise pass under finite representability, so ℓ_p would have cotype s . This contradicts the fact that the infimal cotype exponent of ℓ_p is p . Finally, ℓ_∞ cannot be finitely representable in X , because X has (for example) cotype 3, while ℓ_∞ has no finite cotype. Thus no exponent other than 2 belongs to $S(X)$.

Johnson–Szankowski's cited example is nonreflexive. Since every superreflexive space is reflexive, it is non-superreflexive, completing the identification. $\square$

Scope note 1. *This settles only Tokarev's Problem 2. The HAP question remains genuinely stronger. Johnson–Szankowski ask whether every HAP space is reflexive and explicitly state that this problem remains open [2, p. 1988].*

Verification and novelty status. The implication from the near-Hilbert type/cotype profile to $S(X) = \{2\}$ is an elementary finite-representability argument. The existence input is from the cited literature, so this packet is classified as a *literature-implied answer*, not as a new construction. Cheap run indexes contained no packet for arXiv:math/0206108, and bounded searches for the HAP/nonreflexivity question found no later exact primary-source solution.

References

- [1] E. Tokarev, *An example of a Banach Space with a Subsymmetric Basis, which has the Hereditarily Approximation Property*, arXiv:math/0206108 (2002).
- [2] W. B. Johnson and A. Szankowski, *Hereditary approximation property*, Ann. of Math. (2) **176** (2012), 1987–2001.
- [3] G. Pisier and Q. H. Xu, *Random series in the real interpolation spaces between the spaces v_p* , in *Geometrical Aspects of Functional Analysis (1985/86)*, Lecture Notes in Math. 1267, Springer, 1987, 185–209.